

Coursework Summary

For topic selection discussion, compulsory 4th-semester project

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I'm currently in my Master's in Mathematics at Monash University. Below is what I've studied across my first three semesters. I've noted typical topics for each unit, happy to send the official unit guide for any of these if useful.

Semester 1 (S1 2025)

Unit	Title	Typical topics
MTH3251/4251	Financial mathematics	Stochastic calculus basics, Black-Scholes option pricing, interest rate models, portfolio theory, derivatives pricing
MTH3011/4015	Partial differential equations	Classification of PDEs (elliptic/parabolic/hyperbolic), heat/wave/Laplace equations, separation of variables, Fourier series methods
MTH3320/4320	Computational linear algebra	Numerical methods for linear systems, matrix decompositions (LU, QR, SVD), eigenvalue algorithms, iterative methods, numerical stability
MTH3330/4330	Optimisation and operations research	Linear programming, simplex method, integer programming, nonlinear optimisation, network flows

Semester 2 (S2 2025)

Unit	Title	Typical topics
MTH3230/4230	Time series and random processes in linear systems	Stationary processes, ARMA models, spectral analysis, forecasting, state-space models and the Kalman filter (taught by Dr Tianhai Tian)
MTH3060/4060	Advanced ordinary differential equations	Existence/uniqueness theorems, systems of ODEs, stability theory, phase plane analysis, boundary value problems
MTH2137/3137	Number theory and cryptography	Divisibility, modular arithmetic, congruences, RSA and public-key cryptosystems, primality testing
MTH3020/3025/4020	Complex analysis and integral transforms	Analytic functions, Cauchy-Riemann equations, contour integration, residue calculus, Laplace/Fourier transforms

Semester 3 (S1 2026)

Unit	Title	Typical topics
MTH4089/5540	Statistical learning in finance	Statistical/machine learning methods applied to financial data, regression, classification, risk and asset modelling
MTH5153	Combinatorics	Counting principles, permutations/combinations, graph theory basics, generating functions
MTH5311	Methods of applied mathematics	Asymptotic methods, perturbation theory, integral equations, special functions, mathematical modelling techniques
FIT5147	Data exploration and visualisation	Exploratory data analysis, statistical graphics, data visualisation tools, storytelling with data

For Context, the Project Itself

- Compulsory project for my 4th semester, topic is entirely up to me.
- Current supervisor: Dr Tianhai Tian (School of Mathematics, Monash), taught me MTH3230.
- Areas I'm leaning toward: time series / statistics applied to finance and economics, real-world forecasting and risk problems.
- Two papers he shared with me for inspiration (both his own recent work): deep learning uncertainty/interval prediction for multivariate time series, and graph neural networks for spatiotemporal forecasting.

Note: topic descriptions above reflect standard content for units with these names/codes, based on general knowledge of the field, not scraped Monash handbook text (their handbook pages are JavaScript-rendered). Worth a quick cross-check against the official unit guide on handbook.monash.edu where precision matters.